

On Quantum Algorithms for Weyl Systems, Modular Subset Sum Problem Type Problems, and an Architectural Complexity Separation with an Application of Architectural Complexity Separation

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Abstract

We study a YES/NO promise problem on Weyl systems of $2^n - 1$ genuine equations encoded into n qubit rotations. YES: every equation's modular error is within a fixed tolerance ϵ . NO: one qubit is within tolerance and the other $n - 1$ carry Weyl-distributed error bounded below by a fixed constant. We show two independent mechanisms, the Wobbly QFT and the Latent QFT, recover the anchor period with explicit, ϵ -quantified probability in the YES case, at every finite n , and with probability $\rightarrow 0$ as $n \rightarrow \infty$ in the NO case, under one equidistribution assumption per mechanism, both stated and used only within this paper. No classical hardness is claimed: a classical sampler distinguishes the two cases with a single random query, error probability $2^{-(n-1)}$. The paper is self-contained: all cited results are proved in the appendices.

1 Setup

Fix $n \geq 1$, $N := 2^n$, non-degenerate algebraically independent irrationals $\gamma_1, \dots, \gamma_n$, a bijection $\text{Permute} : \{0, \dots, N - 1\} \rightarrow \{0, 1\}^n$, and write $I(X, z)$ for bit z of $\text{Permute}(X)$. For a candidate integer vector $\mathbf{c} = (c_1, \dots, c_n)$ and targets α_X , the Weyl system is the collection of equations

$$\delta_X(\mathbf{c}) := \left(\sum_{z=1}^n c_z \gamma_z I(X, z) - \alpha_X \right) \bmod 1 = \sum_{z: I(X, z)=1} \eta_z \pmod{1}, \quad X = 0, \dots, N - 1,$$

folded to $(-\frac{1}{2}, \frac{1}{2}]$, where η_z denotes qubit z 's own error contribution (called A_z for the Wobbly QFT and ε_z for the Latent QFT below). Fix the correct solution $\mathbf{c}^*$, so α_X is defined such that $\delta_X(\mathbf{c}^*) = 0$ for every X .

Remark 1 (The system has $N - 1$, not N , genuine equations). *Since Permute is a bijection, there is a unique X_0 with $\text{Permute}(X_0) = 0^n$, so $I(X_0, z) = 0$ for every z . Then $\delta_{X_0}(\mathbf{c}) = (0 - \alpha_{X_0}) \bmod 1$ for every candidate $\mathbf{c}$, and since α_{X_0} is itself defined from the correct solution's empty sum ($\alpha_{X_0} = (\sum_z c_z^* \gamma_z I(X_0, z)) \bmod 1 = 0$), $\delta_{X_0}(\mathbf{c}) = 0$ identically, for every candidate, including maximally wrong ones. Equation X_0 carries no information about $\mathbf{c}$ and is not a genuine constraint. The Weyl system therefore comprises $N - 1 = 2^n - 1$ substantive equations (an odd number, one less than a power of two). This does not affect any bound below, all of which are either exact finite- N statements unaffected by a single fixed equation, or asymptotic statements in n where the difference between N and $N - 1$ is immaterial.*

Both mechanisms act on the periodic anchor $|\psi\rangle = \frac{1}{\sqrt{A}} \sum_{l=0}^{A-1} |x_0 + lr\rangle$, where $p > 3$ is prime, b satisfies $1 < b < p$, $r := \text{ord}_p(b)$ is odd, $n = 2\lceil \log_2 p \rceil$, $N = 2^n$, $A \approx N/r$, and the good set is $G_r := \{y \in \{0, \dots, N-1\} : |y/N - k/r| < 1/(2r^2) \text{ for some integer } k\}$ (Legendre's theorem: such y yield $k/\text{gcd}(k, r)$ over $r/\text{gcd}(k, r)$ via continued fractions). The anchor period is known before it is recovered and a continued fraction algorithm is used to recover the anchor period ; there is a similar process in Shor's algorithm.

2 The YES/NO Promise Problem

Definition 1 (YES/NO promise, with tolerance). *Fix a distinguished index z_0 (which represents the single "clean" qubit) and a tolerance $\epsilon > 0$.*

YES $\sum_{z=1}^n |\eta_z| \leq \epsilon$.

NO $|\eta_{z_0}| \leq \epsilon$, and $|\eta_z| \geq \epsilon_{\text{noise}}$ for every $z \neq z_0$, for a fixed constant $\epsilon_{\text{noise}} > \epsilon$, generic and Weyl-distributed.

Remark 2 (The tolerance controls every equation). *Since $\delta_X(\mathbf{c}) = \sum_{z:I(X,z)=1} \eta_z$, the triangle inequality gives $\max_X |\delta_X(\mathbf{c})| \leq \sum_z |\eta_z|$, so the YES condition guarantees $|\delta_X(\mathbf{c})| \leq \epsilon$ for every one of the $N-1$ genuine equations (Remark 1). Since $\sum_z \eta_z^2 \leq (\sum_z |\eta_z|)^2$, the same ϵ bound controls both threshold quantities used below: $S_n := \sum_z \eta_z^2 \leq \epsilon^2$ for the Wobbly QFT, and $\eta := \sum_z |\eta_z| \leq \epsilon$ itself for the Latent QFT.*

3 Algorithm 1: The Wobbly QFT

Qubit z 's phase error is $A_z := \eta_z$, entering the noisy QFT via $2\pi j/2^z \mapsto 2\pi j/2^z - A_z$ (redrawn independently per branch of $|\psi\rangle$; full definitions in Appendix A).

Theorem 1 (YES case). *If $\sum_z |A_z| \leq \epsilon$, then, unconditionally, for every n ,*

$$P(y \in G_r) \geq \frac{4}{\pi^2} e^{-0.4\epsilon^2} > 0.$$

Proof. By Remark 2, $S_n \leq \epsilon^2$; apply Theorem 8. □

Theorem 2 (NO case). *Under Assumption 1, since $|A_z| \geq \epsilon_{\text{noise}}$ for each of the $n-1$ qubits $z \neq z_0$, $S_n \geq (n-1)\epsilon_{\text{noise}}^2 \rightarrow \infty$ as $n \rightarrow \infty$, and by Theorem 9, $P(y \in G_r) \rightarrow 0$.*

Remark 3. *Qubit z_0 's bounded error contributes at most ϵ^2 to S_n , a fixed amount that never prevents $S_n \rightarrow \infty$; it is the $n-1$ growing, individually bounded-below terms that force divergence. Numerically, with $A_z = 1$ for all $n-1$ noisy qubits, $P(y \in G_r)$ falls from 0.35 at $n = 8$ to 0.13 at $n = 14$, heading toward the floor $|G_r|/N$, consistent with $S_n = n-1 \rightarrow \infty$ and not a step-function collapse at any fixed n .*

4 Algorithm 2: The Latent QFT

Qubit z 's deviation from its target rotation is $\varepsilon_z := \eta_z$ (full definitions in Appendix A). With $\Delta(j) := \sum_{z:j_z=1} \varepsilon_z$, the exact amplitude (Theorem 7) is $\alpha_y = \frac{1}{\sqrt{AN}} \sum_l e^{i\Delta(x_0+lr)} e^{-2\pi i(x_0+lr)(y-1)/N}$.

Theorem 3 (YES case). *If $\eta := \sum_z |\varepsilon_z| \leq \epsilon < 0.0927$, then, unconditionally, for every finite n ,*

$$P(y - 1 \in G_r) \geq \frac{4}{\pi^2} - 4\epsilon - 4\epsilon^2 > 0.$$

Proof. Apply Theorem 10 with the aggregate bound $\eta \leq \epsilon$. □

Theorem 4 (NO case). *Under Assumption 2, since $|\varepsilon_z| \geq \epsilon_{\text{noise}}$ for each of the $n - 1$ noisy qubits, $\eta_n \not\rightarrow 0$ as $n \rightarrow \infty$, and by Theorem 11, $P(y - 1 \in G_r) \rightarrow 0$.*

Remark 4. *As in Algorithm 1, qubit z_0 's bounded contribution to η is a fixed amount that never prevents $\eta_n \not\rightarrow 0$ once the $n - 1$ noisy terms, each bounded below, dominate.*

5 Classical Baseline

Theorem 5 (Trivial classical solvability). *A single uniformly random query among the $N - 1$ genuine equations (Remark 1) distinguishes YES from NO with error probability $2^{-(n-1)}$.*

Proof. $\delta_X(\mathbf{c}) = \sum_{z: I(X,z)=1} \eta_z$ can only appear to satisfy a tolerance well below ϵ_{noise} in the NO case if every noisy $z \neq z_0$ has $I(X, z) = 0$ (genericity of the noisy η_z excludes systematic cancellation among any active noisy terms). Exactly $2^{-(n-1)}$ of all N indices X satisfy this (each of the $n - 1$ noisy bits independently 0 with probability $\frac{1}{2}$ under a generic permutation); this count includes the single degenerate X_0 of Remark 1, which is uninformative in both YES and NO alike and so does not change the error probability. A query landing on any of the other $N - 2^{-(n-1)}N$ indices correctly detects NO. □

6 Discussion

Both mechanisms follow the same shape, governed entirely by the appendix's general threshold: an explicit, ϵ -quantified success probability whenever the aggregate error is bounded ($S_n \leq \epsilon^2$ or $\eta \leq \epsilon$, Theorems 8 and 10), and probability $\rightarrow 0$ once a growing number of individually bounded-below error terms accumulate ($S_n \rightarrow \infty$ or $\eta_n \not\rightarrow 0$, Theorems 9 and 11). A single corrupted qubit, however large its own error, contributes only a fixed, bounded amount to either threshold quantity and so can never by itself force the divergence that drives the NO case to zero; divergence requires a growing count of comparably-corrupted qubits, which is exactly the regime this paper's NO case realizes with $n - 1$ of them.

The classical baseline is trivial and unaffected by the choice between exact and tolerance-bounded cleanliness, since it depends only on which bits are active in a query, never on the size of any qubit's error. No complexity claim is made or needed for the mathematical content to be of interest.

While the classical baseline for this specific YES/NO promise problem is trivial, unaffected by the exact size of any qubit's error since it depends only on which bits are active in a query, the structural mechanics of the quantum embedding offer a compelling pathway for exploring complexity-theoretic separations, particularly between MA and QCMA. The Wobbly and Latent QFTs verify $2^n - 1$ equations simultaneously in polynomial time, relying on global phase interference rather than local sampling. If the NO condition were modified such that the Weyl-distributed noise was sparse or adversarially correlated, effectively hiding from classical random sampling, a classical verifier would require exponential time to query the system. In contrast, the global amplitude collapse governed

by the aggregate thresholds suggests the quantum verifier could still detect the NO case in polynomial time. Thus, this algebraic phase-embedding framework serves as a promising foundation for future work constructing standard oracle separations or custom oracle separations (e.g. in the ACS framework where such a separation is based on the classical intractability of certain problems and is thus a unrelativised white box verification type problem causing such a separation here) between classical and quantum verification classes. No complexity claim is needed for the mathematical content of this exact paper to be of interest, but the architecture itself demonstrates how quantum interference natively processes algebraic constraints.

Appendix A

A Exact Output Distributions

A.1 The Wobbly QFT

Fix per-qubit widths $\varepsilon_1, \dots, \varepsilon_n \geq 0$ matching the realized $|A_z|$. On every application of the noisy QFT to branch l of $|\psi\rangle$, for every qubit z , draw i.i.d. $A_z^{(l)} \sim \text{Unif}[-\varepsilon_z, \varepsilon_z]$ (fresh for each pair (z, l)) and replace $2\pi j/2^z \mapsto 2\pi j/2^z - A_z^{(l)}$ on that qubit; write α_y^ε for the resulting random amplitude and $P(y) := \mathbb{E}[|\alpha_y^\varepsilon|^2]$.

Theorem 6 (Exact formula, Wobbly QFT). *For every $y \in \{0, \dots, N-1\}$,*

$$P(y) = \frac{1}{N} + \left(\prod_{z:y_z=1} \text{sinc}(A_z)^2 \right) \left(P_{\text{ideal}}(y) - \frac{1}{N} \right), \quad \text{sinc}(x) := \frac{\sin x}{x}.$$

Proof. Write $\alpha_y^\varepsilon = \frac{1}{\sqrt{AN}} \sum_l e^{2\pi i(x_0+lr)y/N} e^{-i \sum_{z:y_z=1} A_z^{(l)}}$, so $|\alpha_y^\varepsilon|^2 = \frac{1}{AN} \sum_{l,l'} (\dots)$. The A diagonal terms ($l = l'$) have the random phase cancel exactly, contributing $A \cdot \frac{1}{AN} = \frac{1}{N}$. Each of the $A(A-1)$ off-diagonal terms carries two independent random phases, contributing $\mathbb{E}[e^{-iA_z^{(l)}}] \mathbb{E}[e^{iA_z^{(l')}}] = \text{sinc}(\varepsilon_z)^2$ per bit with $y_z = 1$, reproducing $P_{\text{ideal}}(y) - \frac{1}{N}$ scaled by $\prod_{z:y_z=1} \text{sinc}(A_z)^2$. $\square$

A.2 The Latent QFT

Fix $\varepsilon_1, \dots, \varepsilon_n$ matching the realized η_z ; these are drawn once per trial and applied coherently to every branch of $|\psi\rangle$ (unlike the Wobbly QFT's per-branch redraw above). Qubit z 's rotation is $\theta_z := 2\pi/2^z + \varepsilon_z$. The mask $M(j) := \sum_{z:j_z=1} \theta_z$ is applied to the anchor, $|\psi_M\rangle := \frac{1}{\sqrt{A}} \sum_l e^{iM(x_0+lr)} |x_0 + lr\rangle$, followed by a clean inverse QFT: $\alpha_y := \langle y | \text{QFT}^{-1} | \psi_M \rangle$, $P(y) := |\alpha_y|^2$.

Theorem 7 (Exact formula, Latent QFT). *With $\Delta(j) := \sum_{z:j_z=1} \varepsilon_z$,*

$$\alpha_y = \frac{1}{\sqrt{AN}} \sum_{l=0}^{A-1} e^{i\Delta(x_0+lr)} e^{-2\pi i(x_0+lr)(y-1)/N}.$$

Proof. Since $\sum_{z:j_z=1} \frac{1}{2^z} = j/N$, $M(j) = \frac{2\pi j}{N} + \Delta(j)$. The coefficient of $|j_l\rangle$ in $|\psi_M\rangle$ is $e^{i(2\pi j_l/N + \Delta(j_l))}$, and applying QFT^{-1} ($\langle y | \text{QFT}^{-1} | j \rangle = \frac{1}{\sqrt{N}} e^{-2\pi i j y / N}$) gives $\alpha_y = \frac{1}{\sqrt{AN}} \sum_l e^{i\Delta(j_l)} e^{2\pi i j_l / N} e^{-2\pi i j_l y / N} = \frac{1}{\sqrt{AN}} \sum_l e^{i\Delta(j_l)} e^{-2\pi i j_l (y-1) / N}$. $\square$

Appendix B

B The General Threshold

Lemma 1 (Sinc bound). $\text{sinc}(x)^2 \geq e^{-0.4x^2}$ for $x \in [0, 1]$.

B.1 Wobbly QFT threshold

Theorem 8 (Sufficiency). $S_n := \sum_z A_z^2 = O(1)$ implies, for every n and every assignment of (A_z) realizing S_n , $P(y \in G_r) \geq \frac{4}{\pi^2} e^{-0.4S_n} > 0$.

Proof. Since $\text{sinc}(A_z)^2 \leq 1$, restricting the product of Theorem 6 to a subset of indices can only increase it, so $\prod_{z: y_z=1} \text{sinc}(A_z)^2 \geq \prod_{z=1}^n \text{sinc}(A_z)^2 =: \Sigma_n$ for every y . Summing Theorem 6 over G_r : $P(y \in G_r) = \frac{|G_r|}{N} (1 - \Sigma_n) + \Sigma_n P_{\text{ideal}}(G_r) \geq \Sigma_n \cdot \frac{4}{\pi^2}$ (both terms nonnegative), and $\Sigma_n \geq e^{-0.4S_n}$ by Lemma 1. $\square$

Assumption 1 (Bit equidistribution). There is $c_0 \in (0, \frac{1}{2}]$, independent of n, p, r , with $\rho_z(r) := \Pr_{y \sim G_r}[y_z = 1] \in [c_0, 1 - c_0]$ for every z , and the product-domination bound $\mathbb{E}_{y \sim G_r}[\prod_{z: y_z=1} (1 - u_z)] \leq \prod_z (1 - c_0 u_z)$ for all $u_z \in [0, 1]$.

Theorem 9 (Necessity). Under Assumption 1, if $S_n \rightarrow \infty$ as $n \rightarrow \infty$, then $P(y \in G_r) \rightarrow 0$.

Proof. Apply Assumption 1 with $u_z = 1 - \text{sinc}(A_z)^2$: $\mathbb{E}_{y \sim G_r}[\prod_{z: y_z=1} \text{sinc}(A_z)^2] \leq \prod_z (1 - c_0(1 - \text{sinc}(A_z)^2)) \leq \exp(-c_0 \sum_z (1 - \text{sinc}(A_z)^2)) \leq e^{-0.29c_0 S_n}$, using $1 - x \leq e^{-x}$ and Lemma 1. Together with $|G_r|/N \rightarrow 0$ (since $|G_r| = O(r) = O(p)$ while $N = \Theta(p^2)$), Theorem 6 gives $P(y \in G_r) \rightarrow 0$. $\square$

B.2 Latent QFT threshold

Theorem 10 (Sufficiency). For $\eta := \sum_z |\varepsilon_z| < 0.0927$, unconditionally, for every finite n , $P(y-1 \in G_r) \geq \frac{4}{\pi^2} - 4\eta - 4\eta^2 > 0$.

Proof. Using $|G_r| \leq 2r$ and $A \leq 2N/r$: from Theorem 7 and Cauchy-Schwarz, $|\alpha_y - \alpha_{y-1}^{\text{ideal}}| \leq \eta \sqrt{A/N}$, so $||\alpha_y|^2 - |\alpha_{y-1}^{\text{ideal}}|^2| \leq |\alpha_y - \alpha_{y-1}^{\text{ideal}}| (|\alpha_y - \alpha_{y-1}^{\text{ideal}}| + 2|\alpha_{y-1}^{\text{ideal}}|)$. Summing over the (at most $2r$) points of G_r , with $\sum_{y-1 \in G_r} |\alpha_{y-1}^{\text{ideal}}| \leq \sqrt{|G_r|} \sqrt{P_{\text{ideal}}(G_r)} \leq \sqrt{2r}$ (Cauchy-Schwarz, $P_{\text{ideal}}(G_r) \leq 1$): $\sum |P(y) - P_{\text{ideal}}(y-1)| \leq |G_r| \eta^2 (A/N) + 2\eta \sqrt{A/N} \sqrt{2r} \leq 4\eta^2 + 4\eta$. Combined with $P_{\text{ideal}}(G_r) \geq 4/\pi^2$, the claim follows. $\square$

Assumption 2 (Weyl pseudo-randomness). There is $c_0 > 0$ with $|\mathbb{E}[e^{i(\Delta(j_l) - \Delta(j_{l'}))}]]| \leq 1 - c_0$ for $l \neq l'$, under genericity of the noisy ε_z .

Theorem 11 (Necessity). Under Assumption 2, if $\eta_n \not\rightarrow 0$ as $n \rightarrow \infty$, then $P(y-1 \in nG_r) \rightarrow 0$.

Proof. By Theorem 7 and Assumption 2, the off-diagonal ($l \neq l'$) contribution to $\mathbb{E}[|\alpha_y|^2]$ is damped by a factor bounded away from 1 uniformly. Combined with $|G_r|/N \rightarrow 0$, the argument of Theorem 9 carries over verbatim to give $P(y-1 \in G_r) \rightarrow 0$. $\square$